

ITSOLERA PVT LTD

AI & Data Science – Internship Tasks

Due Date: 20 March 2026

Overview

As part of your AI & Data Science Internship at IT SOLERA PVT LTD, you are required to complete all 4 tasks listed below within the given timeline.

These tasks are designed to provide practical exposure to real-world data science problems, including **ML optimization, feature engineering and model evaluation**.

You are encouraged to complete all tasks for deeper learning and a stronger portfolio. You will work with popular libraries such as **pandas, numpy, scikit-learn, xgboost, shap, matplotlib, streamlit, seaborn, and joblib/pickle**.

Advanced Task Set

Task 1: End-to-End ML Project with Streamlit GUI

Objective:

Build a complete ML system with an interactive GUI for predictions.

Dataset:

Any previously used dataset (e.g., Customer Churn, Loan Prediction, Insurance, Credit Risk)

Instructions:

- Load and preprocess the dataset
- Train multiple models: Logistic Regression, Random Forest, XGBoost
- Save the best-performing model using joblib/pickle
- Build a **Streamlit GUI** with:
 - Input fields for user data
 - Prediction button showing result and probability
 - Visual outputs: Feature importance, dataset preview, model accuracy

Skills Gained:

- End-to-end ML pipeline
 - GUI-based AI application
 - Model integration with frontend
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Task 2: Recommendation System with Streamlit GUI

Objective:

Build a recommendation system with GUI for items or movies.

Dataset:

MovieLens, Netflix, or Amazon dataset

Instructions:

- Build **content-based filtering** model using cosine similarity
- Streamlit GUI:
 - User inputs a movie/product
 - System recommends **top 5 similar items** with similarity score

Skills Gained:

- Recommendation systems
 - Similarity algorithms
 - GUI integration
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Task 3: Model Optimization and Evaluation

Objective:

Improve model performance and systematically evaluate multiple models.

Dataset:

Any previously used dataset

Instructions:

- Train at least 3 models: Logistic Regression, Random Forest, XGBoost/CatBoost
- Apply **hyperparameter tuning** (GridSearchCV or RandomizedSearchCV)
- Evaluate and compare models using:
 - Accuracy, Precision, Recall, F1-Score, ROC-AUC
- Present comparison **in table or chart** (no GUI required)
- Select the best model and justify

Skills Gained:

- Hyperparameter tuning
 - Model evaluation metrics understanding
 - Selecting best model for production
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Task 4: Feature Engineering Challenge

Objective:

Improve model performance through creating and selecting better features.

Dataset:

Any previously used dataset

Instructions:

- Engineer new features:
 - Polynomial features
 - Interaction features
 - Binning/categorization
- Select important features using:
 - Feature importance from tree-based models
 - Correlation analysis
 - Recursive Feature Elimination (RFE)
- Train the best model on the **new feature set**
- Compare performance **before and after feature engineering**

Skills Gained:

- Feature creation & selection
 - Improving model performance
 - Data-driven ML optimization
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Submission Requirements (Per Task)

Each completed task must be uploaded to your **GitHub repository** and shared via **Google Classroom**.

Checklist:

1. **Jupyter Notebook**

- Problem Statement and Objective
 - Dataset description and loading
 - Data cleaning and preprocessing
 - Exploratory Data Analysis (EDA)
 - Model building and evaluation
 - Visualizations (charts, plots, graphs)
 - Final conclusion with insights
 - 2. **Code Quality**
 - Well-structured, readable, and commented code
 - 3. **GitHub Repository**
 - Clear and descriptive repository name
 - README.md including:
 - Task objective
 - Your approach
 - Results and findings
 - 4. **Submission on Google Classroom**
 - Share the link to your GitHub repository after completing each task
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Important Notes

- Complete **4 out of 4 tasks** by **20 March 2026**
 - Ask mentors for help or feedback whenever needed
 - Showcase your work on LinkedIn or GitHub for better visibility
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Task Presentation Requirement

- After completing each task, interns are required to present their work:
 - Short presentation (10–15 minutes)
 - Explain problem statement, approach, key visualizations, results, and insights
 - Presentations will be conducted online
 - Interns must be able to clearly explain their own code

Note: Tasks without presentation will be considered incomplete. Missing even one presentation may affect internship completion.